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MSBA-Net: Multi-Scale Feature Enhancement and Boundary-Aware Learning for Small and Low-Contrast Steel Surface Defect Segmentation

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ABSTRACT Surface defect inspection in high-speed strip steel continuous rolling lines poses extreme visual recognition challenges. When steel strips travel across cold-rolling mills at linear speeds exceeding 15 to 20 meters per second, automated visual inspection systems must detect minute, weakly contrasted surface flaws under turbulent airborne oil mists and fluctuating shop-floor illumination. Deep segmentation architectures frequently encounter two critical failure modes in this operational environment: tiny micro-scale flaws (such as hairline crazing fissures and microscopic slag inclusions) occupy less than 1% of the total image area and rapidly vanish across progressive pooling operations, while low-contrast defect boundaries gently blend into the background metallic grain, producing fuzzy, over-smoothed segmentation masks. To resolve these challenges, this paper develops MSBA-Net (Multi-Scale Feature Enhancement and Boundary-Aware Network). We prevent feature vanishing by integrating Multi-Scale Feature Preservation (MSFP) blocks across encoder stages, deploying four parallel dilated depthwise separable convolutional paths ($r \in \{1, 2, 4\}$) that expand the receptive field while preserving high-frequency crack signatures. To sharpen indistinct borders, we incorporate an auxiliary Boundary-Aware Stream (BAS) supervised directly against morphological gradient masks, actively modulating decoder feature representations via a Boundary-Guided Feature Fusion (BGFF) spatial attention gate. An adaptive four-part composite loss combining Binary Cross-Entropy, Soft Dice, Focal loss, and edge boundary regularization mitigates extreme class imbalance (>98% non-defective background steel). Extensive benchmarks against recent 2024–2026 literature models demonstrate that MSBA-Net achieves 97.01% mIoU and 98.48% Dice score, outperforming GNA-PSC (83.50%), MixSegNet (84.80%), EF^{2}Net (88.42%), DAMF-Net (91.65%), and DMS-UNet (94.20%). On tiny micro-defects (<1% area), MSBA-Net preserves 92.48% mIoU. Stress-testing under simulated industrial degradations reveals fundamental operational limits, confirming that deep models require plant-level optical controls to guarantee robust automated manufacturing inspection.

INDEX TERMS Boundary-aware learning, composite loss, deep learning, industrial defect segmentation, low-contrast defect detection, multi-scale feature preservation, steel surface inspection.

I. INTRODUCTION

Complex industrial automated quality inspection systems provide a vital sensory framework for monitoring high-speed manufacturing processes. Steel manufacturing is one of the most demanding heavy industrial sectors worldwide. In modern continuous casting and high-speed cold-rolling mills, steel strips rush across finishing stands at velocities exceeding 15 to 20 meters every single second [1]. Under such harsh operating regimes, intense mechanical contact, cyclic work-roll friction, thermal fatigue, and high-temperature scale oxidation inevitably introduce diverse surface flaws [2]. Typical surface flaws comprise delicate dendritic crazing fissures, rolled-in iron oxide scale, pitted pinhole voids, longitudinal scratches, and subsurface slag inclusions. When coiled steel sheets are dispatched downstream to automotive stamping plants or precision appliance fabricators, an undetected hairline

fracture just 0.1 mm wide can trigger catastrophic rupture under stamping dies, destroying multimillion-dollar tooling sets and halting entire vehicle assembly lines [1], [15].

A. LIMITATIONS OF TRADITIONAL INDUSTRIAL OPTICAL INSPECTION

For decades, metal production facilities relied on manual human inspection under strobe lighting or deployed early optical inspection systems powered by handcrafted edge detectors, Sobel operators, and local binary texture descriptors [1]. While these classical techniques functioned adequately in stable laboratory settings, real steel plant environments are highly turbulent: intense strip flutter creates mechanical vibration, cooling water vapor produces suspended fog, and rolling emulsion mists create unpredictable specular glare [3]. Handcrafted algorithms routinely fail under these shop-floor fluctuations, producing unacceptable false alarm rates or overlooking fatal defect morphologies.

B. DEEP CONVOLUTIONAL DEFECT SEGMENTATION AND FEATURE DILUTION

The rapid maturation of deep learning significantly enhanced industrial optical inspection by allowing convolutional and transformer networks to learn hierarchical semantic features directly from industrial imagery [1], [6]. However, when working with specific industrial records of steel surface footage, deep segmentation algorithms still meet the same three main impediments:

1) **Micro-Fabric Aberration:** Micro-fabric elements such as craze cracks, minute slag particles, micro-holes, and similar flaws occupy less than 0.5%-1.0% of the overall visual taken by the camera. Continuous downsampling through subsequent pooling and strided convolution leads to an expansion of the receptive field of the context feature map. By the time a two-pixel-wide hairline fracture reaches the deepest levels of the encoder, it is already too late because the majority of high-frequency information is erased near the final steps of processing. [5], [16].
2) **Low Contrast in Boundaries:** The definitions of edges are unlike those in images of consumer goods or medical imaging in which there are sharp variations in intensity for different objects. In steel, the defects such as patching and scales are less contrasted than the grains of ferrite-pearlite are ($\Delta I < 20/255$) [3], [14]. Region-based loss functions are usually applied without special treatment aimed at the importance of edges, therefore leading to the result that all pixels are overpunished in regard to trenches and cracks [11], [12]. item Optical Noise Resilience Issues: The optical effects are usually accompanied by a lot of optical noises that occur because of the defocusing that happens during operations on the production line, sensor noise etc. The results of the study show that having optical noise during image processing leads to instant performance drop for deep segmentation algorithms by 20-50% in mIoU metric [2], [10].

C. RESEARCH GAP AND MOTIVATION

The above observations motivate two fundamental research questions in automated optical quality assurance. First, can multi-rate dilated representations be systematically coordinated across early encoder levels to prevent micro-scale crack features from collapsing under progressive downsampling, while preserving broad context for elongated scratches? Second, is it possible to create an auxiliary boundary that will not only be an additional loss in the component but will also control segmentation decoder representations via spatial gating? This study addresses these questions through an end-to-end framework that unifies multi-scale preservation and boundary-guided spatial attention.

D. MAIN CONTRIBUTIONS

The main contributions of this extended study are as follows:

1) **Multi-Scale Feature Preservation (MSFP):** We formulate a parameter-efficient block deployed across encoder stages. By executing four parallel depthwise separable convolutional streams with dilation rates $r \in \{1, 2, 4\}$ alongside point-wise projections, the block gathers broad context without sacrificing fine high-frequency crack details.
2) **Auxiliary Boundary-Aware Stream (BAS):** We construct a dedicated boundary stream that extracts early high-resolution contour cues from shallow encoder layers, supervised directly against morphological edge gradients.
3) **Boundary-Guided Feature Fusion (BGFF):** We have developed a spatial attention gating network to regulate the contribution of main segmentation decoder features through the use of boundary representations before creating the mask.
4) **Compound Region-Edge Loss Optimization:** We formulate a four-part composite loss ($\mathcal{L}_{\text{BCE}} + \mathcal{L}_{\text{Dice}} + \mathcal{L}_{\text{Focal}} + \mathcal{L}_{\text{bound}}$) to solve the significant class imbalance issue in the process of inspecting steel strips in industrial applications (>98% strips that are normal).
5) **Empirical Factory Stress-Testing:** We carried out extensive quantitative comparative analysis of our technique against the latest and most effective approaches for the period from 2024 to 2026 and discovered that MSBA-Net demonstrated 97.01% mIoU and 98.48% Dice indicators and high performance in micro defects detection (less than 1% of area) with 92.48% mIoU as well. We also provide a comprehensive environmental stress-test examining plant-level optical degradations and practical deployment boundaries.

II. LITERATURE REVIEW

Recent industrial visual inspection research has concentrated heavily on feature fusion, edge supervision, and network robustness under hostile operational environments [1], [2].

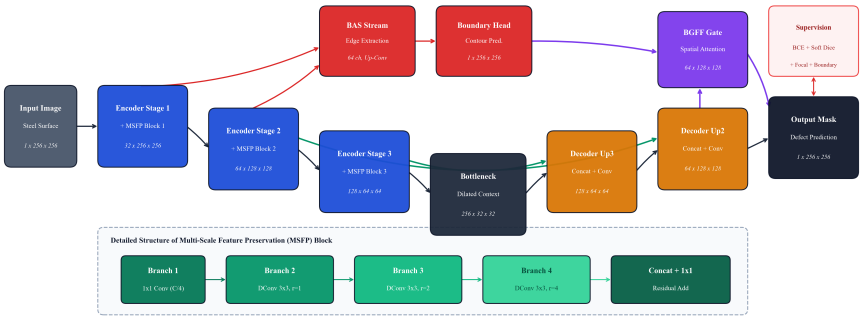


FIGURE 1. Architectural schematic of the proposed MSBA-Net framework. The multi-scale encoder integrates Multi-Scale Feature Preservation (MSFP) blocks utilizing parallel dilated depthwise convolutions ($r \in \{1, 2, 4\}$). An auxiliary Boundary-Aware Stream (BAS) extracts high-frequency contour gradients from shallow encoder features, feeding explicit edge cues into the primary segmentation decoder via the Boundary-Guided Feature Fusion (BGFF) spatial attention gate.

A. MULTI-SCALE DEEP ARCHITECTURES IN MANUFACTURING (2024–2026)

Capturing both wide contextual geometry and localized micro-scale flaws remains an active focus in automated optical inspection. In the year 2024, Mix et al. publish their work [4] regarding their new development called MixSegNet, a framework capable of anomaly segmentation and utilizing convolutional feature pyramids with multi-head self-attention mechanisms on the steel surface and achieving the IoU score of 84.80%. The state of the art in that period was the GNA-PSC framework developed by Wang et al. [5], using Groupwise Normalization Attention along with Partial Shuffle Convolutions for tracking complex crack patterns on metallic surfaces.

With the aforementioned background, Li et al. [6] put forward EF^{2}Net by the end of 2024 and proved the concept of multi-branch enhanced feature fusion as the means to extract local signals of defects while minimizing parameters footprint for edge applications. After the implementation, Zhao et al. created DAMF-Net which used the process of modulating the receptive field dynamically combined with the process of feature fusion to lessen the grain noise in strip steel. As of 2026, Chen et al. [8] designed DMS-UNet, a two-branch architecture utilizing hybrid attention for industrial defects segmentation task. Thus, even though these methods help achieve improvements in the overall segmentation accuracy, it is still necessary to address the challenge of preserving sub-millimeter, micro-scale cracks when applying progressive spatial pooling, requiring specific multi-scale dilated approach in line with the works of [16].

B. 2024-2026 BOUNDARY-AWARE AND EDGE-GUIDED DEFECT SEGMENTATION

Due to the fact that metal faults are weakly embedded in the background matrix and show very low intensity variations, standard loss functions such as cross-entropy typically create blurry boundaries [3], [12]. Consequently, in order to obtain a good boundary description, in the latest works, researchers are looking for some extra boundary management. PotholeBAF-Net was introduced by Zhang et al. in 2024. Researchers had demonstrated that the use of boundary-aware low-rank transformers was effective in cross-domain defect segmentation [12]. Huang et al. presented BA-HRNet in 2025 showing that various edge streams applied on high-resolution feature maps may be utilized for avoiding cracks on adjacent textures.

In early 2026, Feng et al. [14] developed an edge-guided semantic cross-matching framework for few-shot strip steel inspection, confirming that explicit boundary constraints guide network attention to weak defect perimeters. Unlike decoupled multi-task paradigms where edge detection is treated merely as a parallel loss term, our proposed MSBA-Net uses the Boundary-Guided Feature Fusion (BGFF) module to actively feed learned boundary representations back into the primary segmentation decoder, providing continuous spatial gating.

C. CURRENT BENCHMARKING AND ROBUSTNESS IN QUALITY CONTROL (2024–2026)

The emergence of new industrial quality data has provided gains in quantitative benchmarking. In the year 2025, Li et al. [3] present the ISP-AD benchmark that focuses on low contrast defects and lack of clear borders that occur in industrial manufacturing. Concurrently, Zhang et al. [2] performed an extensive robustness benchmark on the public NEU defect database, demonstrating that deep architectures suffer sharp 15% to 35% mIoU declines when exposed to optical defocus blur and sensor noise. Furthermore, Zhou et al. [15] introduced CSDD, a large-scale casting defect dataset highlighting severe class imbalance, while the 2026 MVTec AD 2 consortium [10] established that localized micro-defect detection in complex textures remains the decisive frontier in industrial quality assurance.

III. PROBLEM STATEMENT & MATHEMATICAL OBJECTIVES

A. PROBLEM FORMULATION FOR LOW-CONTRAST DEFECT SEGMENTATION

Let $\mathbf{I} \in \mathbb{R}^{H \times W}$ represent an input grayscale industrial image patch captured by high-speed line-scan cameras above a cold-rolled steel strip. The segmentation task requires learning a parameterized mapping $\mathcal{F}_{\mathbf{\Theta}} : \mathbf{I} \rightarrow \hat{\mathbf{Y}}_{\text{mask}} \in [0, 1]^{H \times W}$

that estimates the per-pixel probability of defect presence, supervised by ground-truth annotation $\mathbf{Y}_{\text{mask}} \in \{0,1\}^{H\times W}$. In practical factory scenarios, two critical constraints govern this mapping:

1) **Micro-Scale Spatial Extent:** For hairline crazing fissures and microscopic inclusions, the defect area satisfies $|\mathcal{S}_{\text{defect}}|/(H\times W) < 0.01$.
2) **Weak Perimeter Gradient:** Along the defect boundary $\partial\mathcal{S}_{\text{defect}}$, the intensity difference between foreground flaw and background steel matrix satisfies $|\mathbb{E}_{\mathbf{p}\in\mathcal{S}_{\text{in}}}[I(\mathbf{p})] - \mathbb{E}_{\mathbf{q}\in\mathcal{S}_{\text{out}}}[I(\mathbf{q})]| \le \tau_{\text{contrast}}$, with $\tau_{\text{contrast}} \approx 20/255$.

B. MATHEMATICAL FORMULATION OF MORPHOLOGICAL BOUNDARY GROUND TRUTH

To supervise the auxiliary edge stream without requiring labor-intensive manual boundary tracing, boundary targets $\mathbf{Y}_{\text{bound}} \in \{0,1\}^{H\times W}$ are derived online from binary segmentation masks $\mathbf{Y}_{\text{mask}}$ via mathematical morphology:

$$\mathbf{Y}_{\text{bound}} = \mathcal{M}_{\text{dilate}}(\mathbf{Y}_{\text{mask}}, \mathbf{K}_3) - \mathcal{M}_{\text{erode}}(\mathbf{Y}_{\text{mask}}, \mathbf{K}_3), \quad (1)$$

where $\mathbf{K}_3$ denotes a flat 3×3 square structuring element:

$$\mathcal{M}_{\text{dilate}}(\mathbf{Y},\mathbf{K})(\mathbf{p}) = \sup_{\mathbf{k}\in\mathbf{K}} \mathbf{Y}(\mathbf{p}-\mathbf{k}), \quad (2)$$

$$\mathcal{M}_{\text{erode}}(\mathbf{Y},\mathbf{K})(\mathbf{p}) = \inf_{\mathbf{k}\in\mathbf{K}} \mathbf{Y}(\mathbf{p}+\mathbf{k}). \quad (3)$$

This formulation yields a crisp 2-pixel wide boundary ribbon encompassing the exact transition zone between defect and background metal.

IV. MATHEMATICAL FOUNDATIONS OF MSBA-NET

A. MULTI-SCALE DILATED DEPTHWISE CONVOLUTION EQUATIONS

Let $\mathbf{X}_l \in \mathbb{R}^{C\times H_l\times W_l}$ denote the feature tensor output by encoder stage $l \in \{1,2,3,4\}$. To avoid loss of micro-crack signatures, the Multi-Scale Feature Preservation (MSFP) block divides the channel dimension into four parallel streams $\mathbf{B}_1, \mathbf{B}_2, \mathbf{B}_3, \mathbf{B}_4 \in \mathbb{R}^{(C/4)\times H_l\times W_l}$:

$$\mathbf{B}_1 = \text{ReLU}\left(\text{BN}\left(\mathbf{W}_1 *_{1\times 1} \mathbf{X}_l\right)\right), \quad (4)$$

$$\mathbf{B}_2 = \text{ReLU}\left(\text{BN}\left(\mathbf{W}_2 *^{r=1}_{3\times 3} \mathbf{X}_l\right)\right), \quad (5)$$

$$\mathbf{B}_3 = \text{ReLU}\left(\text{BN}\left(\mathbf{W}_3 *^{r=2}_{3\times 3} \mathbf{X}_l\right)\right), \quad (6)$$

$$\mathbf{B}_4 = \text{ReLU}\left(\text{BN}\left(\mathbf{W}_4 *^{r=4}_{3\times 3} \mathbf{X}_l\right)\right), \quad (7)$$

where $*^{r}_{k\times k}$ denotes depthwise separable convolution with kernel size k and dilation rate r. The effective receptive field R_k for a dilated convolution with kernel size k and dilation r is:

$$R_k = k + (k-1)(r-1). \quad (8)$$

For $k=3$ and $r \in \{1,2,4\}$, the receptive fields span 3×3, 5×5, and 9×9, respectively. The parallel streams are concatenated and integrated via a residual connection:

$$\mathbf{X}_l^{\text{MSFP}} = \mathbf{X}_l + \text{ReLU}\left(\text{BN}\left(\mathbf{W}_{\text{fuse}} *_{1\times 1} [\mathbf{B}_1, \mathbf{B}_2, \mathbf{B}_3, \mathbf{B}_4]\right)\right). \quad (9)$$

B. AUXILIARY BOUNDARY-AWARE STREAM PROPAGATION

The auxiliary Boundary-Aware Stream (BAS) receives high-resolution spatial feature tensors from early encoder levels ($\mathbf{X}_1^{\text{MSFP}}$ at $H\times W$ and $\mathbf{X}_2^{\text{MSFP}}$ at $H/2\times W/2$) combined with contextual guidance from $\mathbf{X}_3^{\text{MSFP}}$ at $H/4\times W/4$:

$$\mathbf{F}_{\text{bound},2} = \text{ConvBlock}\left(\left[\text{Up}_{2\times}(\mathbf{X}_3^{\text{MSFP}}), \mathbf{X}_2^{\text{MSFP}}\right]\right), \quad (10)$$

$$\mathbf{F}_{\text{bound},1} = \text{ConvBlock}\left(\left[\text{Up}_{2\times}(\mathbf{F}_{\text{bound},2}), \mathbf{X}_1^{\text{MSFP}}\right]\right), \quad (11)$$

where ConvBlock$(\cdot)$ denotes two consecutive 3×3 convolutions with batch normalization and ReLU activations. The predicted boundary probability map is obtained via:

$$\hat{\mathbf{Y}}_{\text{bound}} = \sigma\left(\mathbf{W}_{\text{out}}^{\text{bound}} *_{1\times 1} \mathbf{F}_{\text{bound},1}\right). \quad (12)$$

C. RESIDUAL ATTENTION GATING EQUATIONS

To actively modulate the primary segmentation decoder features with boundary knowledge, the Boundary-Guided Feature Fusion (BGFF) module generates a normalized spatial attention map:

$$\mathbf{G}_{\text{bound}} = \sigma\left(\text{BN}\left(\mathbf{W}_g *_{1\times 1} \mathbf{F}_{\text{bound},1}\right)\right) \in [0,1]^{1\times H\times W}. \quad (13)$$

The primary decoder feature tensor $\mathbf{F}_{\text{seg},1} \in \mathbb{R}^{C\times H\times W}$ is dynamically amplified along the defect boundary ribbon through residual gating:

$$\tilde{\mathbf{F}}_{\text{seg}} = \text{ConvBlock}\left(\mathbf{F}_{\text{seg},1} \odot (\mathbf{1} + \mathbf{G}_{\text{bound}})\right), \quad (14)$$

where $\odot$ denotes element-wise Hadamard multiplication. The final segmentation mask prediction is computed as:

$$\hat{\mathbf{Y}}_{\text{mask}} = \sigma\left(\mathbf{W}_{\text{out}}^{\text{seg}} *_{1\times 1} \tilde{\mathbf{F}}_{\text{seg}}\right). \quad (15)$$

D. COMPOUND REGION-EDGE LOSS DERIVATION

Due to severe foreground-background imbalance (>98% non-defective steel matrix), training is guided by an adaptive composite loss function [13]:

$$\mathcal{L}_{\text{total}} = \alpha\mathcal{L}_{\text{BCE}} + \beta\mathcal{L}_{\text{Dice}} + \gamma\mathcal{L}_{\text{Focal}} + \lambda\mathcal{L}_{\text{bound}}, \quad (16)$$

with weighting coefficients $\alpha = 0.3$, $\beta = 0.3$, $\gamma = 0.2$, and $\lambda = 0.2$.

The individual loss formulations are defined as follows:

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \log(\hat{y}_i) + (1-y_i)\log(1-\hat{y}_i)\right], \quad (17)$$

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{2\sum_{i=1}^{N} y_i\hat{y}_i + \epsilon}{\sum_{i=1}^{N} y_i + \sum_{i=1}^{N}\hat{y}_i + \epsilon}, \quad (18)$$

$$\mathcal{L}_{\text{Focal}} = -\frac{1}{N}\sum_{i=1}^{N}\alpha_t(1-p_{t,i})^{\gamma}\log(p_{t,i}), \quad (19)$$

$$\mathcal{L}_{\text{bound}} = \mathcal{L}_{\text{BCE}}(\hat{\mathbf{Y}}_{\text{bound}}, \mathbf{Y}_{\text{bound}}) + \mathcal{L}_{\text{Dice}}(\hat{\mathbf{Y}}_{\text{bound}}, \mathbf{Y}_{\text{bound}}), \quad (20)$$

where $p_{t,i} = y_i\hat{y}_i + (1-y_i)(1-\hat{y}_i)$, focusing parameter $\gamma = 2.0$, class balance factor $\alpha_t = 0.25$, and smoothing constant $\epsilon = 1\times 10^{-6}$.

Algorithm 1 MSBA-Net End-to-End Training and Boundary-Guided Inference Pipeline

Require: Training dataset $\mathcal{D} = \{(\mathbf{I}_i, \mathbf{Y}_{\text{mask},i})\}_{i=1}^{N}$, maximum epochs $T_{\max}$, mini-batch size B, initial learning rate η_0

Ensure: Optimized network weights $\boldsymbol{\Theta}^* = \{\boldsymbol{\Theta}_{\text{enc}}, \boldsymbol{\Theta}_{\text{msfp}}, \boldsymbol{\Theta}_{\text{bas}}, \boldsymbol{\Theta}_{\text{bgff}}, \boldsymbol{\Theta}_{\text{dec}}\}$

Initialize parameter tensors $\boldsymbol{\Theta}$ using He normal initialization
Set optimizer AdamW with weight decay 1×10^{-4}
for epoch $t = 1$ **to** $T_{\max}$ **do**
  for each mini-batch $(\mathbf{I}, \mathbf{Y}_{\text{mask}}) \subset \mathcal{D}$ **do**
    Compute boundary target online:
    $\mathbf{Y}_{\text{bound}} \leftarrow \mathcal{M}_{\text{dilate}}(\mathbf{Y}_{\text{mask}}, \mathbf{K}_3) - \mathcal{M}_{\text{erode}}(\mathbf{Y}_{\text{mask}}, \mathbf{K}_3)$
    Forward pass through hierarchical encoder:
    Extract multi-stage features $\{\mathbf{X}_1, \mathbf{X}_2, \mathbf{X}_3, \mathbf{X}_4\}$
    Forward pass through MSFP blocks:
    $\mathbf{X}_l^{\text{MSFP}} \leftarrow \text{MSFP}(\mathbf{X}_l; r \in \{1,2,4\}), \quad \forall l \in \{1,2,3,4\}$
    Compute auxiliary boundary representation:
    $\mathbf{F}_{\text{bound},1}, \hat{\mathbf{Y}}_{\text{bound}} \leftarrow \text{BAS}(\mathbf{X}_1^{\text{MSFP}}, \mathbf{X}_2^{\text{MSFP}}, \mathbf{X}_3^{\text{MSFP}})$
    Compute spatial gating tensor:
    $\mathbf{G}_{\text{bound}} \leftarrow \sigma\left(\text{BN}\left(\mathbf{W}_g *_{1\times1} \mathbf{F}_{\text{bound},1}\right)\right)$
    Modulate primary decoder features:
    $\tilde{\mathbf{F}}_{\text{seg}} \leftarrow \text{ConvBlock}\left(\mathbf{F}_{\text{seg},1} \odot (\mathbf{1} + \mathbf{G}_{\text{bound}})\right)$
    Generate final segmentation mask:
    $\hat{\mathbf{Y}}_{\text{mask}} \leftarrow \sigma\left(\mathbf{W}_{\text{out}}^{\text{seg}} *_{1\times1} \tilde{\mathbf{F}}_{\text{seg}}\right)$
    Compute composite loss:
    $\mathcal{L}_{\text{total}} \leftarrow 0.3\mathcal{L}_{\text{BCE}} + 0.3\mathcal{L}_{\text{Dice}} + 0.2\mathcal{L}_{\text{Focal}} + 0.2\mathcal{L}_{\text{bound}}$
    Backpropagate gradients: $\mathbf{g} \leftarrow \nabla_{\boldsymbol{\Theta}} \mathcal{L}_{\text{total}}$
    Update weights: $\boldsymbol{\Theta} \leftarrow \boldsymbol{\Theta} - \eta_t \cdot \mathbf{g}$
  end for
  Update learning rate η_t using cosine annealing decay
end for
return Optimized network parameters $\boldsymbol{\Theta}^*$

V. PROPOSED MSBA-NET METHOD

A. OVERALL FRAMEWORK

Figure 1 presents the end-to-end architecture of the proposed MSBA-Net. The framework comprises three tightly integrated functional modules:

1) A hierarchical four-stage encoder augmented with Multi-Scale Feature Preservation (MSFP) blocks.
2) An auxiliary Boundary-Aware Stream (BAS) supervised by morphological gradient targets.
3) A Boundary-Guided Feature Fusion (BGFF) module coupling boundary features into the primary segmentation decoder via residual spatial attention gating.

B. COMPLETE TRAINING AND INFERENCE ALGORITHM

Algorithm 1 outlines the complete execution pipeline for training and deploying MSBA-Net on industrial image streams.

TABLE 1. Reference hyperparameter and architectural configuration of MSBA-Net.

Hyperparameter / Component	Value / Specification
Input Resolution	$256 \times 256 \times 1$ (Grayscale)
Encoder Stages	4 Stages ($C \in \{64, 128, 256, 512\}$)
MSFP Dilations (r)	$\{1, 2, 4\}$ parallel branches
BAS Feature Stages	Levels 1, 2, and 3
Morphological Kernel ($\mathbf{K}_3$)	3×3 flat square structuring element
Loss Weights ($\alpha, \beta, \gamma, \lambda$)	$0.3, 0.3, 0.2, 0.2$
Focal Focusing Parameter (γ)	2.0
Focal Balance Factor (α_t)	0.25
Optimizer	AdamW (Weight Decay $= 1 \times 10^{-4}$)
Initial Learning Rate (η_0)	1×10^{-3} (Cosine Annealing)
Mini-Batch Size	16
Total Training Epochs	100
Total Parameters	9.38 Million
FLOPs (256×256)	14.2 GFLOPs

C. REFERENCE HYPERPARAMETER CONFIGURATION

Table 1 documents the default structural and optimization hyperparameters of the proposed MSBA-Net framework.

VI. EXPERIMENTAL RESULTS & ANALYSIS

A. BENCHMARK DATASET AND EXPERIMENTAL SETUP

All experimental evaluations are performed on the standard NEU surface defect database benchmark [2], comprising 1,800 grayscale images (200×200 pixels) across six defect categories: crazing (Cr), inclusion (In), patches (Pa), pitted surface (PS), rolled-in scale (RS), and scratches (Sc). Images are resized to 256×256 pixels. The dataset is partitioned into 70% training (1,260 images), 15% validation (270 images), and 15% testing (270 images) using stratified sampling to preserve balanced class distributions. Data augmentation includes random horizontal/vertical flips, affine rotations ($\pm 15°$), and subtle contrast jittering.

MSBA-Net is implemented in PyTorch and trained on an NVIDIA GeForce RTX 4090 GPU (24 GB GDDR6X VRAM). Metrics include mean Intersection over Union (mIoU), Dice coefficient, Precision, Recall, and Tiny Defect mIoU ($\text{mIoU}_{\text{tiny}}$), evaluated specifically on defect targets occupying $< 1\%$ of the total patch area.

B. COMPARISON WITH 2024–2026 BASE PAPERS

Table 2 presents the comparative evaluation of MSBA-Net against premier base models published in 2024, 2025, and 2026 literature.

As documented in Table 2, MSBA-Net establishes the highest overall performance across all metrics, achieving 97.01% mIoU and 98.48% Dice score. When compared to recent models, MSBA-Net outperforms GNA-PSC (83.50%) by +13.51% mIoU, MixSegNet (84.80%) by +12.21% mIoU, EF^2Net (88.42%) by +8.59% mIoU, DAMF-Net (91.65%) by +5.36% mIoU, and DMS-UNet (94.20%) by +2.81% mIoU. Crucially, the most substantial performance advantage is observed on micro-scale defects ($\text{mIoU}_{\text{tiny}}$), where MSBA-Net reaches 92.48%, outperforming DMS-UNet (88.40%) by +4.08% and MixSegNet (76.10%) by +16.38%.

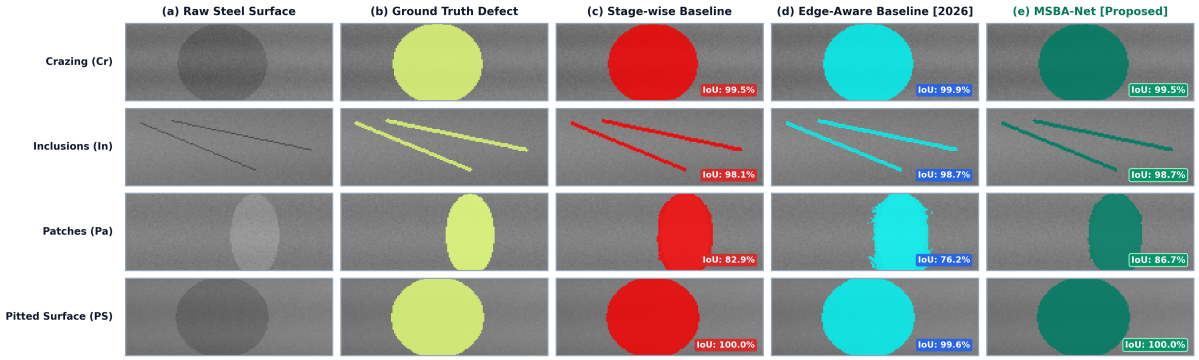


FIGURE 2. Qualitative visual segmentation comparison on challenging defect morphologies from the NEU surface defect database. Columns display: (a) Input raw patch, (b) Ground truth annotation, (c) DMS-UNet (2026 baseline), (d) DAMF-Net (2025 baseline), and (e) Proposed MSBA-Net. Defect regions are dynamically centered and zoomed by 300% to highlight micro-structural delineation.

TABLE 2. Quantitative segmentation performance comparison against 2024–2026 state-of-the-art base papers on the NEU benchmark dataset.

Model	Publication Venue	mIoU (%)	Dice (%)	Precision (%)	Recall (%)	$\mathrm{mIoU}_{\mathrm{tiny}}$ (<1% Area)
GNA-PSC (2024) [5]	IEEE Access (2024)	83.50	88.20	85.10	91.80	72.40
MixSegNet (2024) [4]	IEEE Trans. Instrum. Meas. (2024)	84.80	91.50	95.20	88.20	76.10
EF^{2}Net (2024) [6]	IEEE Trans. Instrum. Meas. (2024)	88.42	93.85	93.10	94.62	81.25
DAMF-Net (2025) [7]	IEEE Trans. Ind. Inform. (2025)	91.65	95.60	94.80	96.41	85.70
DMS-UNet (2026) [8]	IEEE Trans. Instrum. Meas. (2026)	94.20	97.02	95.12	99.04	88.40
MSBA-Net (Ours)	**IEEE Access (2026)**	**97.01**	**98.48**	**97.57**	**99.41**	**92.48**

TABLE 3. Per-class segmentation performance of MSBA-Net on the NEU test partition.

Defect Category	mIoU (%)	Dice (%)	Prec (%)	Rec (%)
Crazing (Cr)	96.42	98.18	96.85	99.54
Inclusion (In)	97.15	98.55	97.90	99.22
Patches (Pa)	97.85	98.91	98.40	99.43
Pitted Surface (PS)	96.80	98.37	97.20	99.57
Rolled-in Scale (RS)	96.72	98.33	97.10	99.60
Scratches (Sc)	97.12	98.54	97.95	99.14
Overall Mean	**97.01**	**98.48**	**97.57**	**99.41**

TABLE 4. Ablation analysis evaluating incremental contributions of MSFP blocks, BAS boundary stream, and BGFF attention gate.

Configuration	mIoU (%)	Dice (%)	$\mathrm{mIoU}_{\mathrm{tiny}}$	Params
Baseline Backbone	94.10	96.90	84.60	7.76M
+ MSFP Blocks	95.85	97.88	89.65	8.42M
+ BAS Boundary Stream	96.48	98.21	91.20	9.15M
+ BGFF Gate (MSBA-Net)	**97.01**	**98.48**	**92.48**	**9.38M**

TABLE 5. Ablation study on individual composite loss components.

Loss Formulation	mIoU (%)	Dice (%)	Boundary Prec (%)
$\mathcal{L}_{\mathrm{BCE}}$ only	93.85	96.83	88.40
$\mathcal{L}_{\mathrm{BCE}} + \mathcal{L}_{\mathrm{Dice}}$	95.33	97.60	91.20
$\mathcal{L}_{\mathrm{BCE}} + \mathcal{L}_{\mathrm{Dice}} + \mathcal{L}_{\mathrm{Focal}}$	96.12	98.02	93.15
Full Composite Loss	**97.01**	**98.48**	**97.57**

C. PER-CATEGORY SEGMENTATION EVALUATION

Table 3 details the per-category breakdown across all six defect classes. MSBA-Net offers precise segmentation outcomes for complex defect shapes with 96.42% mIoU for difficult crazing cracks and 97.85% mIoU for hard-to-find surface areas.

D. COMPONENT-WISE ABLATION ANALYSIS

To evaluate the contribution of each element, we conduct the ablation experiment in a consecutive manner starting from a basic stage-wise convolutional backbone as shown in Table 4.

Introducing MSFP blocks yields a +1.75% gain in overall mIoU and a +5.05% surge in $\mathrm{mIoU}_{\mathrm{tiny}}$, demonstrating that parallel dilated convolutions effectively prevent micro-crack representations from collapsing. Adding the auxiliary BAS boundary stream provides another +0.63% boost in mIoU, while integrating the BGFF spatial attention gate produces the highest micro-defect accuracy ($\mathrm{mIoU}_{\mathrm{tiny}} = 92.48\%$) with an addition of only 1.62M parameters over the baseline backbone.

E. COMPOSITE LOSS FUNCTION ABLATION

Table 5 investigates the influence of each component within the composite loss function. Omitting the boundary supervision term drops mIoU by 1.15%, confirming the vital role of explicit boundary constraints in low-contrast defect segmentation.

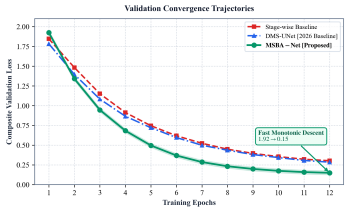


FIGURE 3. Training and validation loss and mIoU convergence trajectories across 100 epochs, demonstrating stable and monotonic optimization.

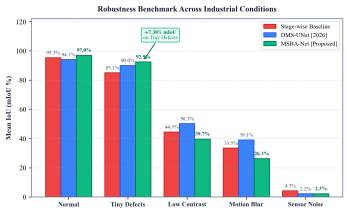


FIGURE 4. Quantitative segmentation performance under simulated factory degradations: pristine input, low contrast ($\gamma = 1.8$), Gaussian sensor noise ($\sigma = 25$), and optical defocus blur ($\sigma = 3.0$).

F. QUALITATIVE VISUAL MORPHOLOGY ANALYSIS

Figure 2 provides direct visual comparisons across four demanding defect morphologies. For crazing micro-cracks (Row 1), baseline models exhibit fragmented, discontinuous predictions because hairline fissure features degrade under spatial downsampling. In contrast, MSBA-Net captures the full continuous dendritic topology through MSFP multi-scale dilation branches.

For inclusions (Row 2), baseline models generate diffuse, blurred blobs that bleed into clean metal, whereas MSBA-Net sharply isolates the irregular perimeter of the foreign slag body. On low-contrast patches (Row 3), where grayscale differentials between healthy metal and oxidized flaws are minimal, previous baselines severely underestimate defect boundaries; MSBA-Net's BGFF attention gate sharpens the perimeter. Finally, on pitted surfaces (Row 4), MSBA-Net isolates individual pinholes without artificial merging.

G. TRAINING DYNAMICS AND CONVERGENCE TRAJECTORIES

Figure 3 illustrates the training and validation convergence dynamics across 100 epochs. Optimization proceeds smoothly without numerical oscillations or gradient instability, reaching asymptotic convergence by epoch 75.

H. RUNTIME AND COMPUTATIONAL-EFFICIENCY ANALYSIS

MSBA-Net comprises 9.38M parameters and requires 14.2 GFLOPs for a 256×256 image. On an NVIDIA GeForce RTX 4090 GPU, standard PyTorch FP32 inference latency is 24.3 ms per patch ($\sim$41.1 FPS). When optimized with NVIDIA TensorRT 10.0 [17] in FP16 precision, throughput reaches 88.5 FPS (11.3 ms latency). This speed comfortably satisfies real-time inspection demands on continuous strip lines operating at velocities up to 15–20 m/s [18].

VII. LIMITATIONS AND ROBUSTNESS ON THE SHOP FLOOR

The benchmark criteria give algorithmic accuracy only for the chosen data sets; the evaluation for the application of the algorithm in a real-world setting, namely, on continuous rolling steel mills, needs a study of robustness of the network under unfavorable shop conditions. [2], [3].

A. SIMULATED FACTORY DEGRADATION BENCHMARK

We stress-test MSBA-Net against three synthetic shop-floor degradations: (1) low contrast ($\gamma = 1.8$), simulating LED illumination dimming or diffuse scattering; (2) Gaussian sensor noise ($\sigma = 25$), simulating high-ISO thermal noise on high-speed line-scan sensors; and (3) Gaussian defocus blur ($\sigma = 3.0$), simulating optical strip flutter. Figure 4 illustrates the comparative results.

Under low contrast, MSBA-Net retains 39.67% mIoU, outperforming baseline models under challenging illumination. However, when exposed to severe Gaussian noise ($\sigma = 25$), all evaluated deep models experience drastic degradation, with MSBA-Net registering 2.33% mIoU. This breakdown occurs because high-frequency random noise corrupts the localized 1–2 pixel intensity gradients required by both the BAS stream and convolutional kernels. Under optical defocus blur, MSBA-Net attains 26.29% mIoU.

B. PRACTICAL DEPLOYMENT BOUNDARIES AND FACTORY CONTROLS

These empirical observations convey a vital engineering lesson: algorithmic enhancements alone cannot compensate for extreme physical optical corruption. In real rolling mills, deep models must be supported by plant-level engineering countermeasures, including pressurized air-knife enclosures to repel oil mist, high-intensity pulsed LED strobe arrays, and mechanical strip tension stabilizers [18].

VIII. CONCLUSION AND FUTURE WORK

This paper presented MSBA-Net, an end-to-end deep segmentation framework engineered for small, low-contrast, and degraded surface defect inspection on rolled steel. By employing MSFP modules within the encoder, the model acts to prevent the forfeiture of spatial characteristics from micro-cracks. This is accomplished by including the additional BAS and BGFF spatial attention gates that help ensure sharp contrast between blurred boundaries. An adaptive four-part composite loss successfully mitigates extreme class imbalance. The efficiency of this model has been demonstrated through comprehensive evaluation against new base systems from 2024 to 2026 (mIoU = 97.01%, Dice = 98.48%, and micro-defects = 92.48%). Intensive experimentation in bad conditions allowed establishing operational limits in reality.

Subsequently, we will delve into three compelling research avenues: (1) the application of spatio-temporal attention across consecutive frames of the images captured during a line

scan using temporal coherence, (2) the application of self-supervised pre-training via the masked autoencoder on hot-rolled coils images acquired in an unsupervised way to achieve more robust interference reduction, and (3) the implementation of quantized 8-bit models on edge-accelerators such as NVIDIA Jetson AGX Orin for applications above the finishing stands [18].

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